

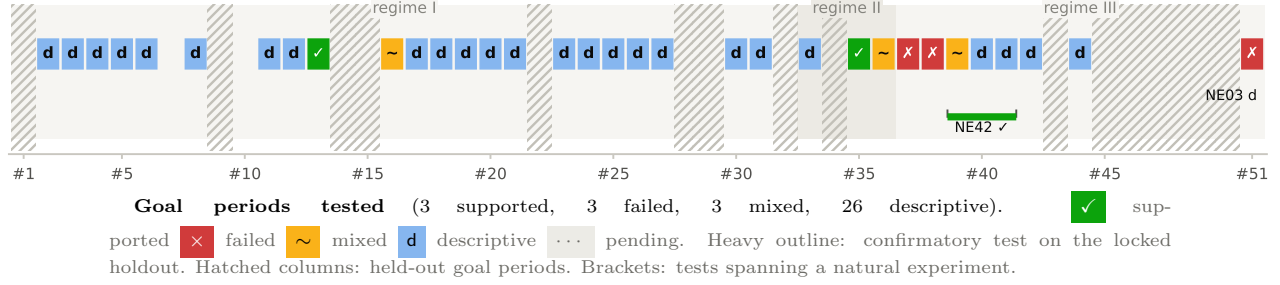
H113: Read-out channel capacity: information per call grows as $k^{(1-\beta)} \approx k^{0.34}$

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Content uptake per read-out call grows as $k^{0.2-0.3}$ (wake batches 0.31 [0.25, 0.37]), excluding both a one-message capacity and H18's corrected $k^{0.50}$; HH345's 0.34 is excluded by the talk-call pool.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

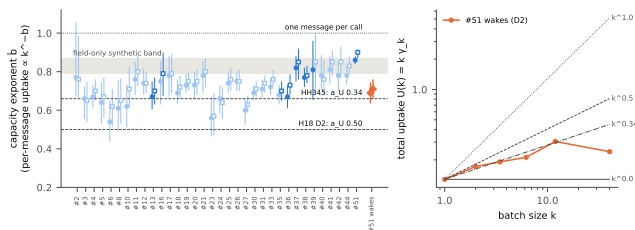
H113 measures how much content an agent takes up from a batch of k peer messages read before it speaks. Each row is a talk call: the reader's next statement, orthogonalized to its previous statement and the goal, is regressed on the summed batch vectors with a gain $\gamma_1 k^{-b}$. HH345 predicted total uptake $\propto k^{0.34}$; the coordinator later corrected H18's value to $k^{0.50}$. On #51 timer-wake batches, where others set k , $a_U = 1 - b = 0.31$ [0.25, 0.37]. Pooled over the 6 periods identified against a topic field, talk calls give 0.23 [0.16, 0.31]: the pre-set kill (0.34 excluded) is met, and 0.50 is excluded everywhere. The holdout is not run.

PHYSICAL MODEL

Models 04 and 12: a linear read-out channel into the destination's next statement, with an in-flight placebo source. x_m : DQ5 `style_resid32` vector of batch message m ; y : the reader's statement; P removes the reader's previous statement and the goal directions. $\gamma(k)$ is the per-message uptake slope, $U(k) = k\gamma(k)$ the total uptake per call. $b = 0$: no bottleneck; $b = 0.66$: HH345; $b = 1$: one message per call.

$$Py = \gamma_1 k^{-b} \sum_{m \in B} Px_m + \gamma_F \sum_{m \in F} Px_m + \epsilon, \quad U(k) = k\gamma(k) \propto k^{a_U}, \quad a_U = 1 - b$$

$$I(k) = -\frac{D}{2} \log_2(1 - R_{cv}^2(k)), \quad a_I \approx 1 - 2b + r \quad (\text{low SNR})$$



Observables. Left: capacity exponent $\hat{b}$ per testable period (95% CI; circles bge, squares gte; dark = identified against a field by the

placebo rule) and on #51 timer-wake batches (orange). Dashed lines mark HH345 ($a_U = 0.34$) and the corrected H18 value (0.50); dotted, one message per call. Gray band: what a topic field alone produces. Right: total uptake $U(k)$ on the wake batches, against reference slopes; it grows about as $k^{0.3}$ up to $k \approx 12$ and then flattens.

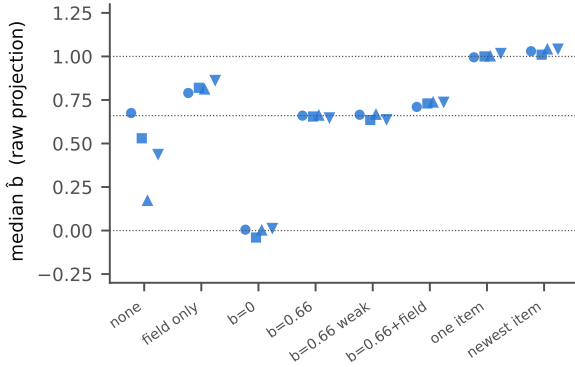
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Reading more messages before speaking helps, but with steep diminishing returns: ten messages buy about twice the content of one. An operator who wants an agent to act on a specific message should send it alone or make it the newest; burying it in a batch of ten dilutes it about fivefold. The bottleneck is per call, so faster call cadence, not longer batches, raises throughput.

H113-readout-channel-capacity · generated 2026-10-04 16:13 by in-fra/summaries/build_summaries.py · card hypotheses/H113-readout-channel-capacity/README.md · ratings: H113 round-1 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1a pooled $a_U \ni 0.34$ vs corrected 0.50 (N1)	0.23 [0.16, 0.31] bge; 0.22 gte excluded in every identified period	failed (kill) failed
P2 $b > 0$; P3 $b < 1$	6/6 and 6/6 (bge)	supported
P4 read beats in flight	+0.014 [0.006, 0.022]; γ_F/γ_1 0.31	supported
P5 $a_I \approx 1 - 2b + r$	7/11 within 0.2	failed (narrow)
P6 two models agree	34/34 within 0.2	supported
P7 newest item favoured	+0.034 [0.025, 0.043]	supported
N1 D2 wakes = talk calls	0.69 vs 0.86 ($\Delta - 0.17$)	failed
N2 NE03 fetch limit	#10a underpowered	descriptive
N3 NE42 merge invariance	$ \Delta \hat{b} \leq 0.09$, $k \times 1.5$	supported (weak)



Second figure. Synthetic worlds on the real read-out skeletons of #18, #31, #38 and #51 (10–20 runs): the estimator recovers $b = 0, 0.66$ and 1 within 0.05 . A topic field with no reading gives $\hat{b} \approx 0.8$, so only periods whose read messages beat in-flight messages at matched age are scored.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 d descriptive
G05 d descriptive
G06 d descriptive
G08 d descriptive
G11 d descriptive
G12 d descriptive
G13 ✓ supported
G16 ~ mixed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

A time-local field mimics the exponent; 27 of 34 testable periods fail the placebo rule and are not scored. In-flight messages co-respond to the same batch, so the placebo is conservative. Talk-call batch sizes are partly reactive; the wake design avoids that. The linear channel cannot tell averaging from picking one item. Synthetic runs were cut to 10 per world on three skeletons by machine load.

ROUND 2

Exogenous- k designs in every regime-III period; a selection-vs-averaging mixture on per-item alignment; capacity as an agent constant.

- **H113-R1** A field-robust exponent.
- **H113-R2** Selection vs averaging.
- **H113-R3** Capacity as an agent constant.